

5 Malaria is an infectious disease caused by the protoctist, *Plasmodium*.

As part of its lifecycle, *Plasmodium* infects human red blood cells. Researchers can compare haemoglobin from the red blood cells of a healthy person with haemoglobin from a person with malaria.

(a) Throughout the world, most deaths from malaria are caused by *P. vivax* and *P. falciparum*.

Name **one** other species of *Plasmodium* that causes malaria.

Plasmodium [1]

(b) In the laboratory, oxygen at different partial pressures can be bubbled through a solution of haemoglobin to determine the percentage saturation of haemoglobin at each partial pressure. A graph constructed from the results is known as an oxygen dissociation curve.

Fig. 5.1 is an oxygen dissociation curve for **normal** adult haemoglobin in **humans**.

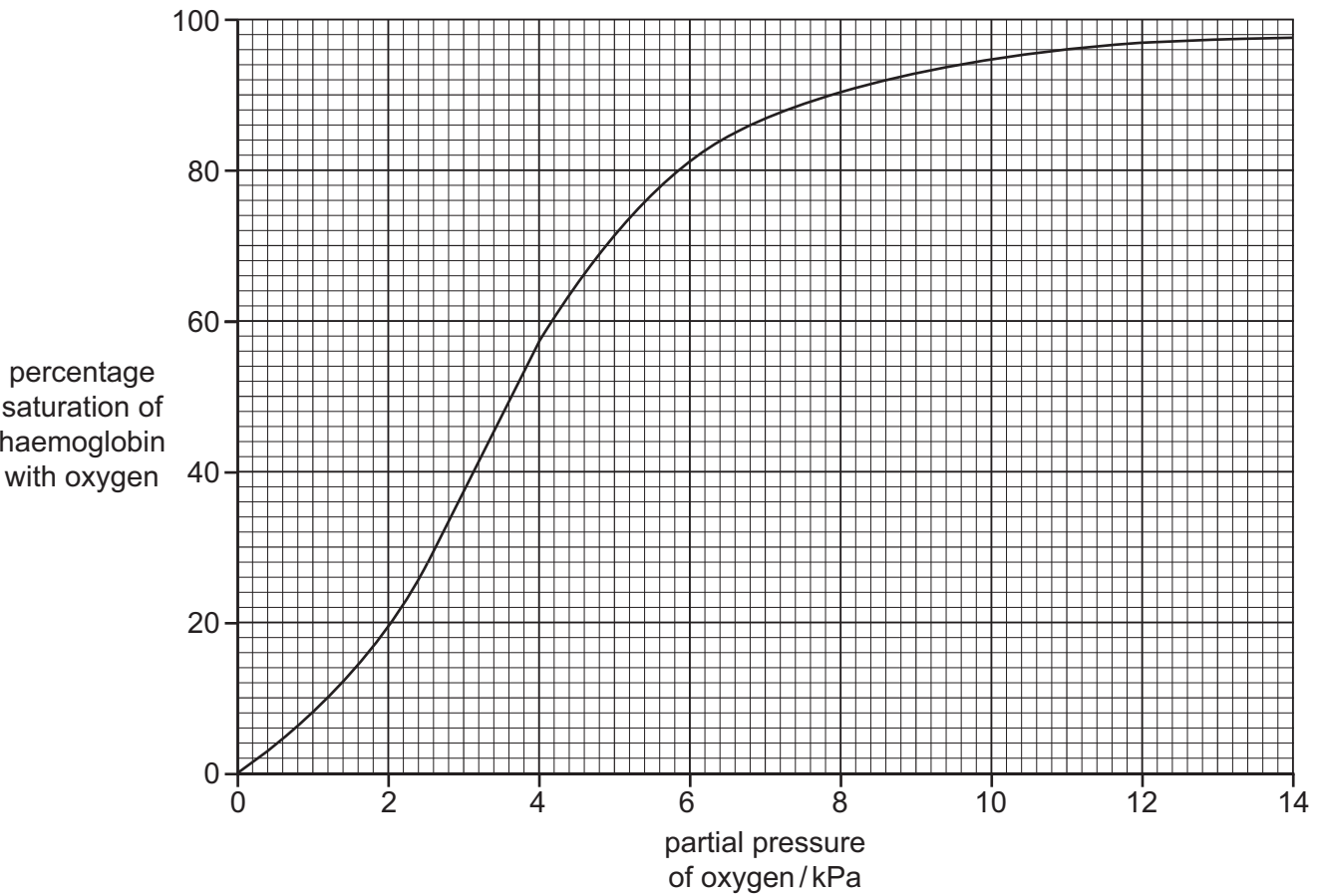
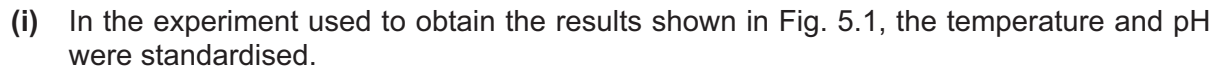


Fig. 5.1



..... [2]

- By analysing the results, the researchers concluded that the oxygen dissociation curve of a person with malaria would be shifted to the right.

..... [3]

- (c) A red blood cell that is infected with *Plasmodium* cannot carry out its function as effectively as a normal red blood cell.

Describe how the size **and** structure of a red blood cell is related to its function, **other than** the fact that it contains a very large number of haemoglobin molecules.

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[Total: 10]